

# Baohua Yang

Principal Architect | Enterprise AI, Agentic Systems & Distributed Platforms  
Bay Area, CA | +1-650-832-5156 | [yangbaohua@gmail.com](mailto:yangbaohua@gmail.com) | [LinkedIn](#) | [Homepage](#)

## PROFILE

---

- Principal Architect and technical leader with 15+ years of research and engineering experience in distributed systems and enterprise platforms, with a current focus on enterprise AI and agentic systems.
- Designs enterprise AI agents and platform capabilities for multi-agent orchestration and context management, combining system architecture with hands-on engineering.
- Open-source maintainer and contributor to Hyperledger and OpenStack; former Hyperledger TSC member; mentor and technical books author.

## TECHNICAL SKILLS

---

- AI: multi-agent orchestration (LangChain/LangGraph), security, RAG, MCP, A2A, context management, evaluation, data protection, inference, fine-tuning, guardrails, OPA.
- Languages & Systems: Python, Go, C/C++, Kubernetes, distributed infrastructure.

## EXPERIENCE

---

### Principal Architect, *Oracle*

Aug 2017 – Present

- Led and built enterprise AI and distributed-system products for Fortune 500 customers, spanning a ~\$200M ARR portfolio and ~50 engineers; set technical direction across AI, security, data, and platform architecture.
- Designed and implemented an enterprise AI agent, improving query-result accuracy from ~30% to 90+% and cutting response latency by 50%. Selected for Oracle's AI Champion program (~100 across the company).
- Designed a supervisor-based multi-agent architecture to replace fixed sequential invocation with dynamic orchestration; built the prototype and drove adoption by the database engineering team.
- Influenced the architecture and feature roadmap of Oracle Database AI capabilities, helping deliver RAG-based querying, context management, and multi-agent orchestration.
- Designed multi-tenant isolation, resource quotas, and autoscaling for Oracle Blockchain Platform across Kubernetes clusters running several thousand pods on containerized Linux infrastructure.
- Solved key technical challenges including production failures, built high-performance tools over 10K TPS, and delivered customer-facing APIs.

### Research Scientist, *IBM*

Jan 2013 – Aug 2017

- Architected and built enterprise blockchain platforms in Go, sustaining over 1,000 TPS per cluster in production, serving hundreds of distributed clusters.
- Core developer on Hyperledger Fabric; drove the applied research that fed the product.

## OPEN SOURCE & COMMUNITY

---

### **Hyperledger Foundation TSC Member (now LF Decentralized Trust) 2017-2019; 2020-2021**

- Helped shape technical strategy and project governance across the Hyperledger ecosystem; contributed to blockchain and distributed ledger standardization through ISO/TC 307.
- Mentored open-source developers and new contributors.

## EDUCATION

---

### **Ph.D. in Control Science and Engineering, *Tsinghua University* Sep 2007 – Jan 2013**

Research focus: High-performance distributed systems and algorithms. 10 Gbps unified threat management (UTM) system.

IBM Ph.D. Fellowship Award (9 winners, 2011).

### **Visiting Scholar, *EECS, University of California, Berkeley*, Sep 2010 – Sep 2011**

Advanced distributed systems, including software-defined and data-center networking.

### **Bachelor of Science and Engineering, *Tsinghua University*, Aug 2003 – Jul 2007**

## SELECTED PATENTS

---

- Methods and System to Support Long Context Management in AI Agentic System, 2026
- Enhanced NL2SQL AI Query System and Method, 2025
- Methods and System to Protect the AI Knowledge Base System, 2024
- [Systems and methods of transaction identification generation for transaction-based environment](#), US 11,822,538 B2 (granted 2023).
- [Methods and system of tracking transactions for distributed ledger](#), US 11,741,075 B2 (granted 2023).
- [Systems and methods of providing ledger as a service](#), US 11,546,425 B2 (granted 2023).
- [Self-adaptive building container images](#), US 10,983,816 B2 (granted 2021).

6 additional granted patents in distributed ledgers, networking, and mobile computing.

## SELECTED PUBLICATIONS

---

- LazyCtrl: A Scalable Hybrid Network Control Plane Design for Cloud Data Centers. K. Zheng, L. Wang, B. Yang, Y. Sun, S. Uhlig. IEEE Trans. Parallel and Distributed Systems, 28(1):115–127, 2017.
- Keep Forwarding: Towards K-link Failure Resilient Routing. B. Yang, J. Liu, S. Shenker, J. Li, K. Zheng. IEEE INFOCOM 2014.
- Practical Multi-tuple Packet Classification using Dynamic Discrete Bit Selection. B. Yang, J. Fong, W. Jiang, Y. Xue, J. Li. IEEE Transactions on Computers, 63(2):424–434, 2014.
- Data-Driven Network Connectivity. J. Liu, B. Yang, S. Shenker, M. Schapira. ACM HotNets-X, 2011.

## SELECTED BOOKS AND TECHNICAL GUIDES

---

- Harness Engineering Guide and Agentic AI Guide (2026)
- Docker Container Technology in Action, China Machine Press, 2015 (4 editions).
- Blockchain: Principle, Design and Applications, China Machine Press, 2017 (2 editions).